

DASHBOARDS FOR DAVIS

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This work presents two interactive dashboard prototypes designed for the touch wall of the Davis visualization environment at TU Wien Library, targeting casual information visualization for academic communities. The prototypes address the challenge of making institutional scientometric data accessible and engaging to non-expert users, including students, researchers and other individuals with personal connections to academic activities.

PROTOTYPE 1 INTERNATIONAL COLLABORATIONS

The "International collaborations" data visualization employs a dual-component architecture capable of handling substantial data volumes while providing intuitive, real-time visual analytics capabilities. The backend datastore manages relational data encompassing institutions, researchers, publications, and geographic distributions. Optimized spatial data structures enable rapid rendering of institution locations and affiliation densities across the global map. The frontend framework adheres to Shneiderman's Mantra: "overview first, zoom and filter, then details-on-demand".

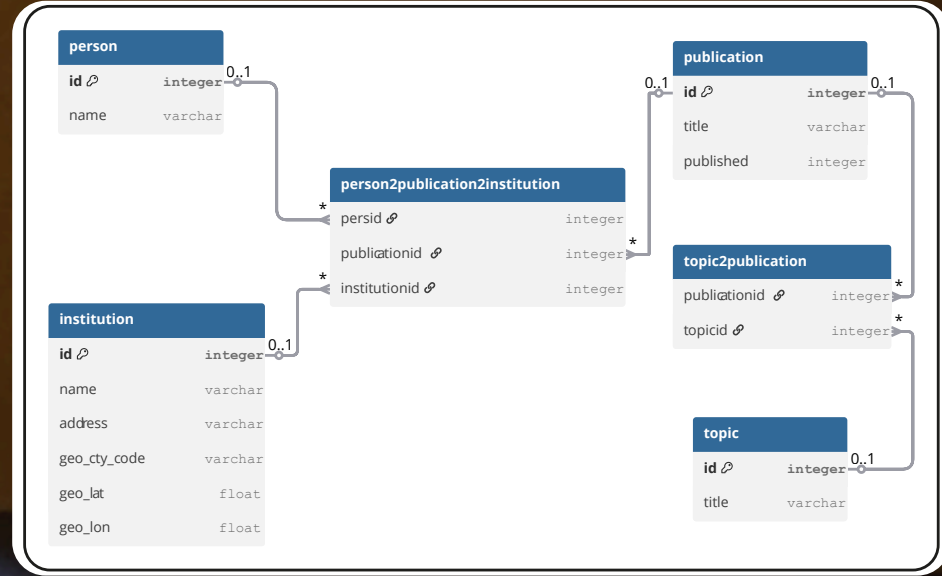


JSON API data

INTRODUCTION

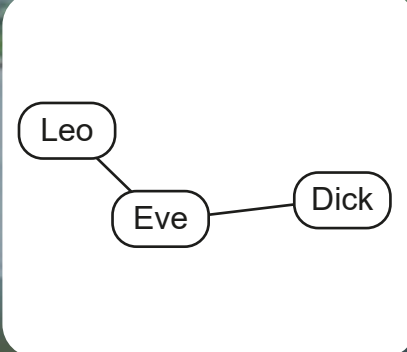
Davis is a visualization environment located at the library of TU Wien, offering visualization infrastructure for teaching and science communication. It provides a 4.6m x 2.6m 4K 3D LED wall for stereoscopic VR visualization and a 4.1m x 1.2m LCD IR-touch wall for interactive presentations. The touch wall is supposed to host an openly accessible digital exhibit that should offer casual users a data-driven "shopwindow" to the activities of the university. Two complementary prototypes have been designed for this purpose, providing insight into TU Wien research and teaching activities through casual information visualization.

Davis
DATA VISUALISATION SPACE

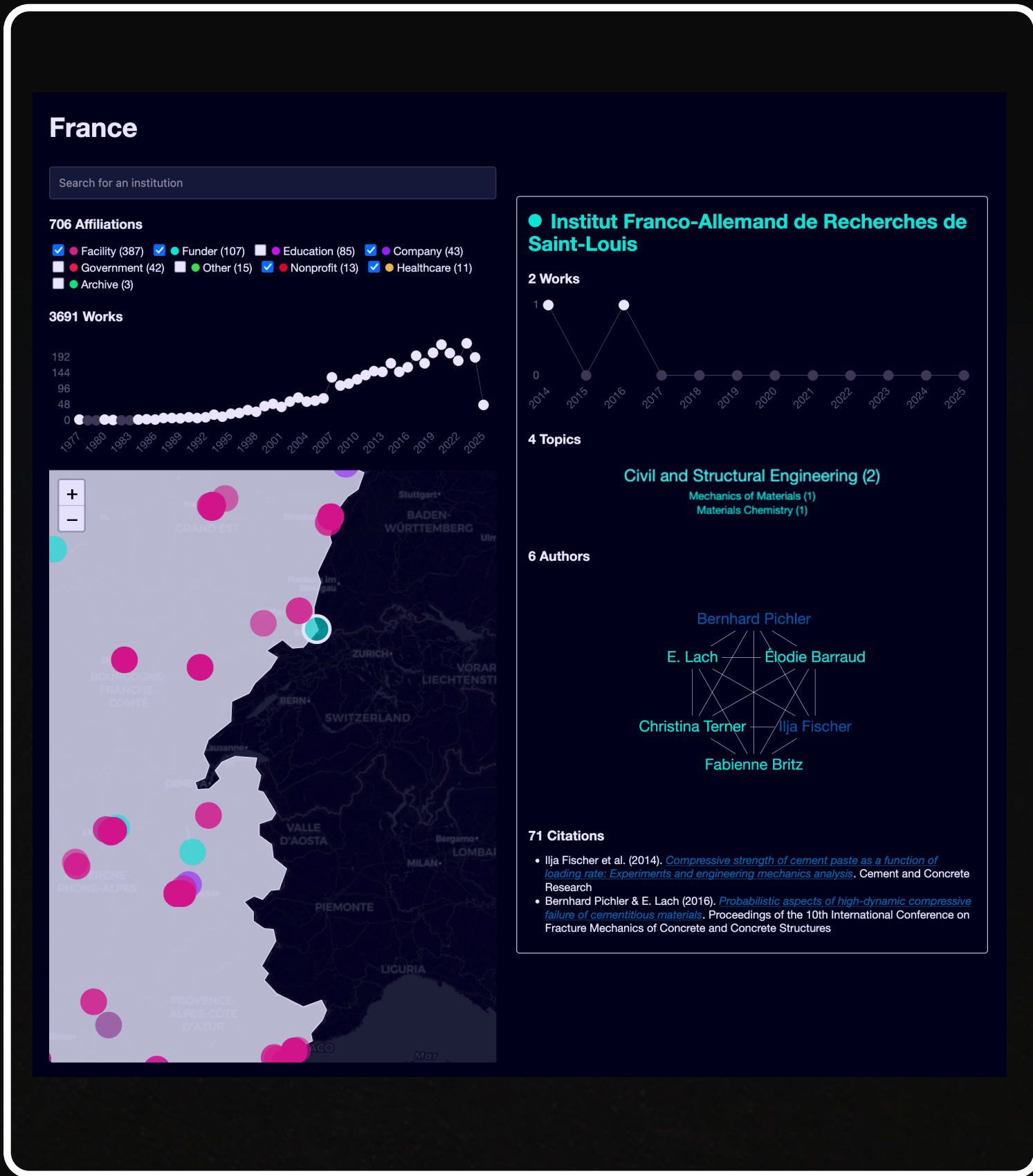
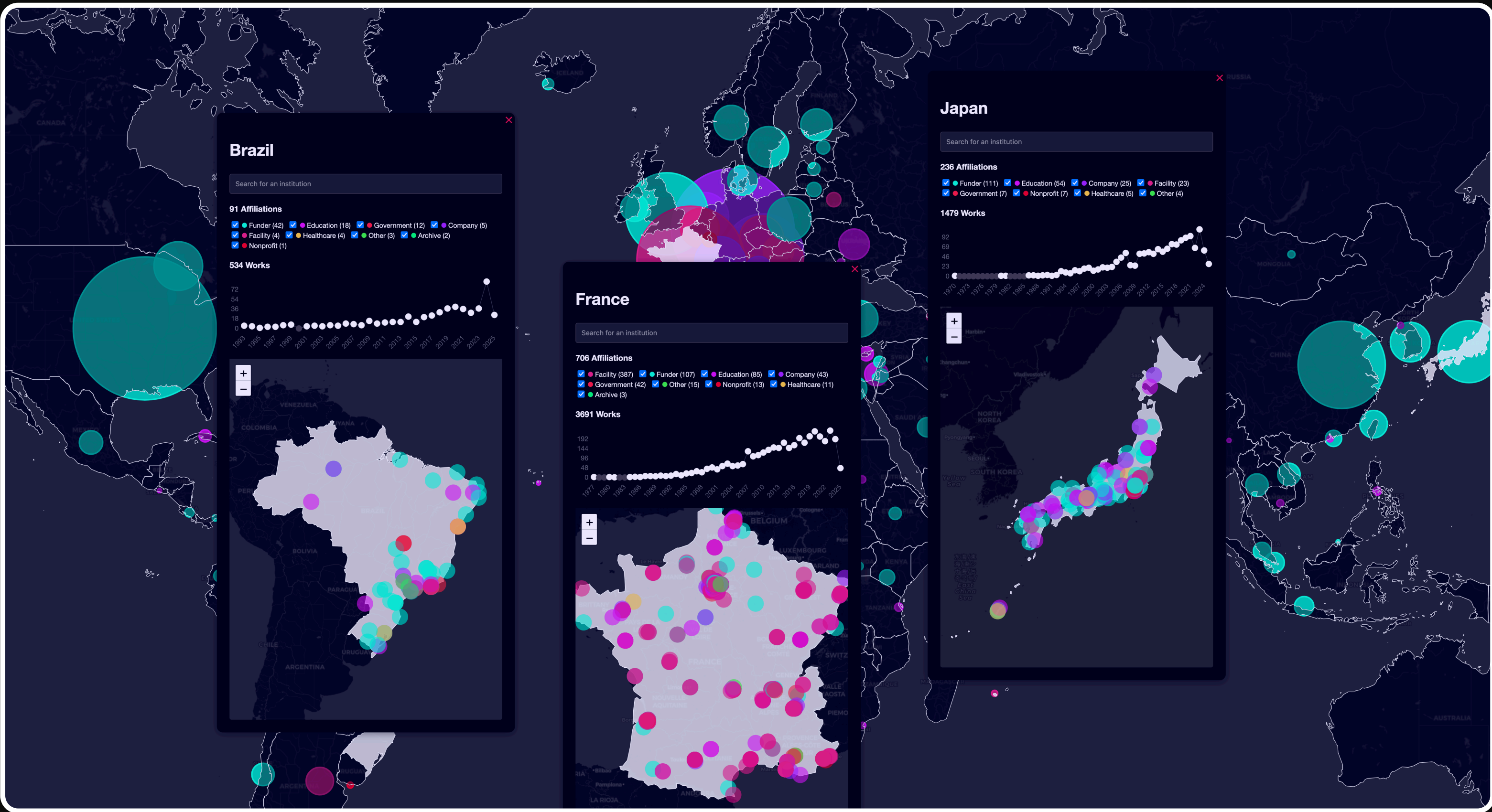


Data Transformation

Person	Pub.-ID	Institution
Alice	1	TU Wien
Bob	1	MIT
Carol	1	TU Graz
Dick	2	TU München
Eve	2	TU Wien
Frank	3	TU Wien
Greta	3	Uni Heidelberg
Heather	4	TU Wien
Ian	4	Uni Milano
Jane	4	Uni Sidney
Karl	4	Uni Heidelberg
Leo	5	TU München
Eve	5	TU Wien



Country & institution data with authors



CONTEXTURAL INFORMATION ARCHITECTURE

The world map serves as the primary navigation interface, leveraging users' inherent spatial cognition to explore landscapes. Interactive selection reveals multi-layered detail panels. Each institution panel displays hierarchical information from broad (country, total affiliations) to specific (individual researchers, recent publications), allowing natural drill-down exploration. Integrated temporal charts provide historical context for institutional research trajectories, supporting both current state assessment and longitudinal analysis.

PROTOTYPE 2 COMMON TOPICS IN STUDENT THESES

Focusing on teaching aspects and student theses in particular, the second prototype seeks to spark interest in teaching and research topics of the university's faculties as they become eminent in the keywords assigned to thesis metadata upon submission. The underlying assumption is that theses and keywords constitute the nodes of a bi-partite network with thesis-keywords edges, yielding a map of interconnected topics. The latter are expected to be clustered by scientific domain but also to feature cross-domain relationships either via common keywords representing general concepts (e.g. "simulation") or via application scenarios where a particular technology or process is applied to another domain.

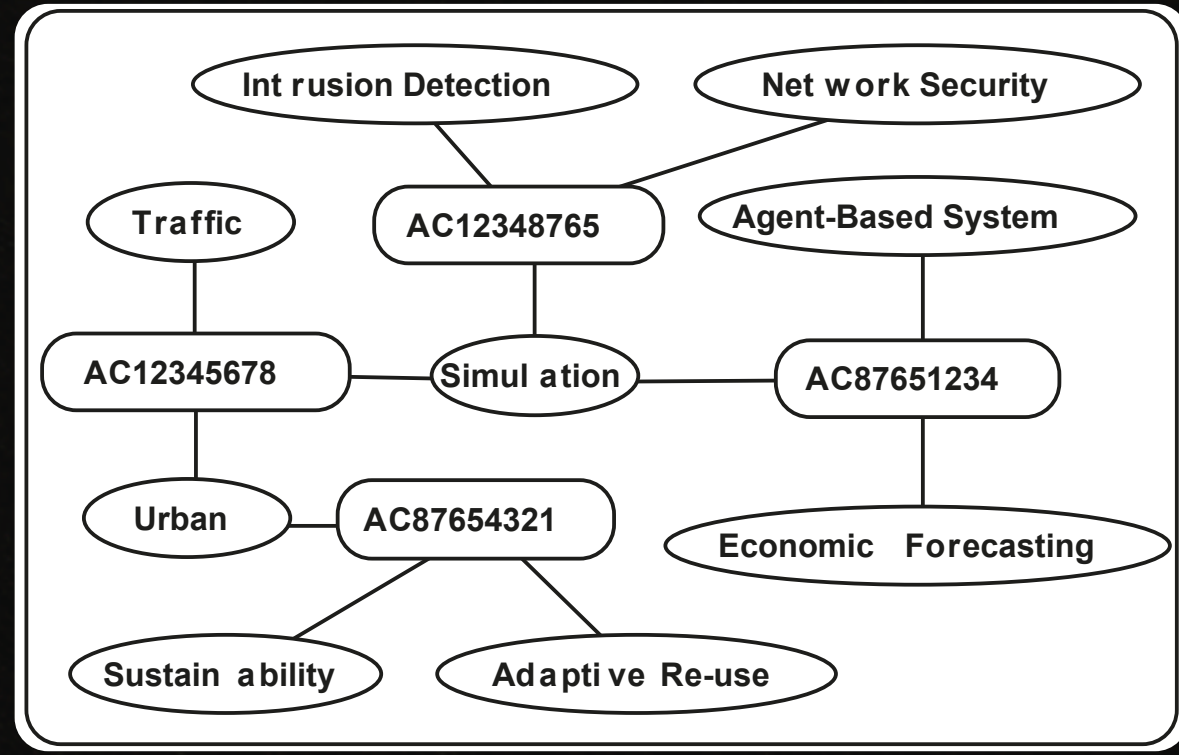


MARC records

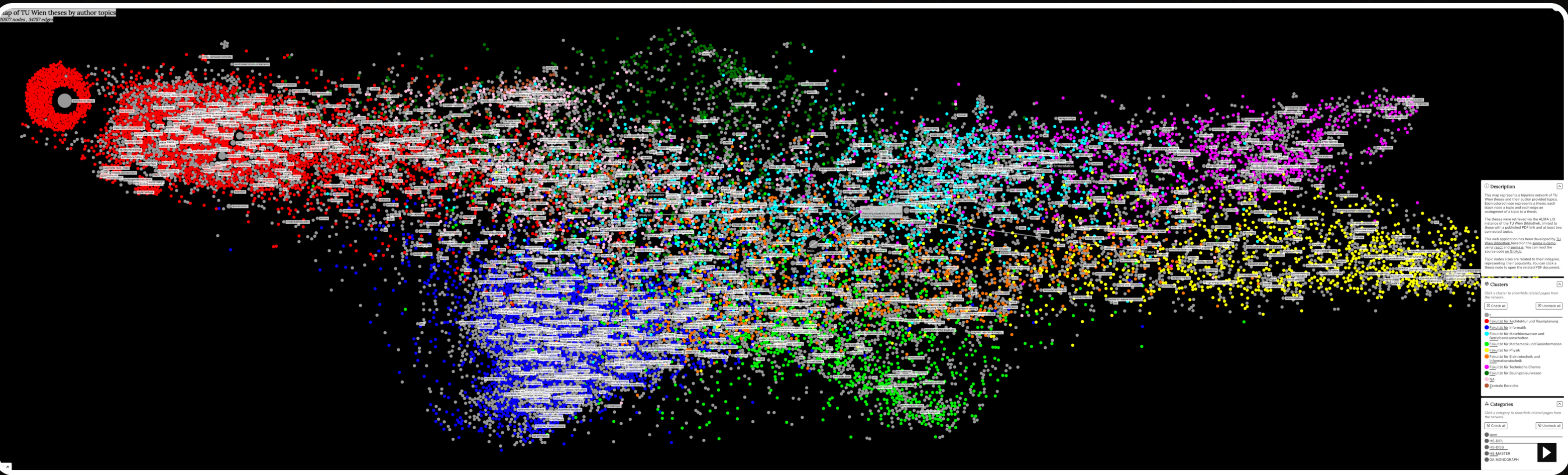
Thesis	Keyword
AC12346578	Simulation
AC12346578	Traffic
AC12346578	Urban
AC87654321	Urban
AC87654321	Sustainability
AC87654321	Adaptive Re-use
AC12348765	Intrusion Detection
AC12348765	Simulation
AC12348765	Network Security
AC87651234	Simulation
AC87651234	Agent-Based System
AC87651234	Economic Forecasting

Keyword Harmonization

Data Transformation

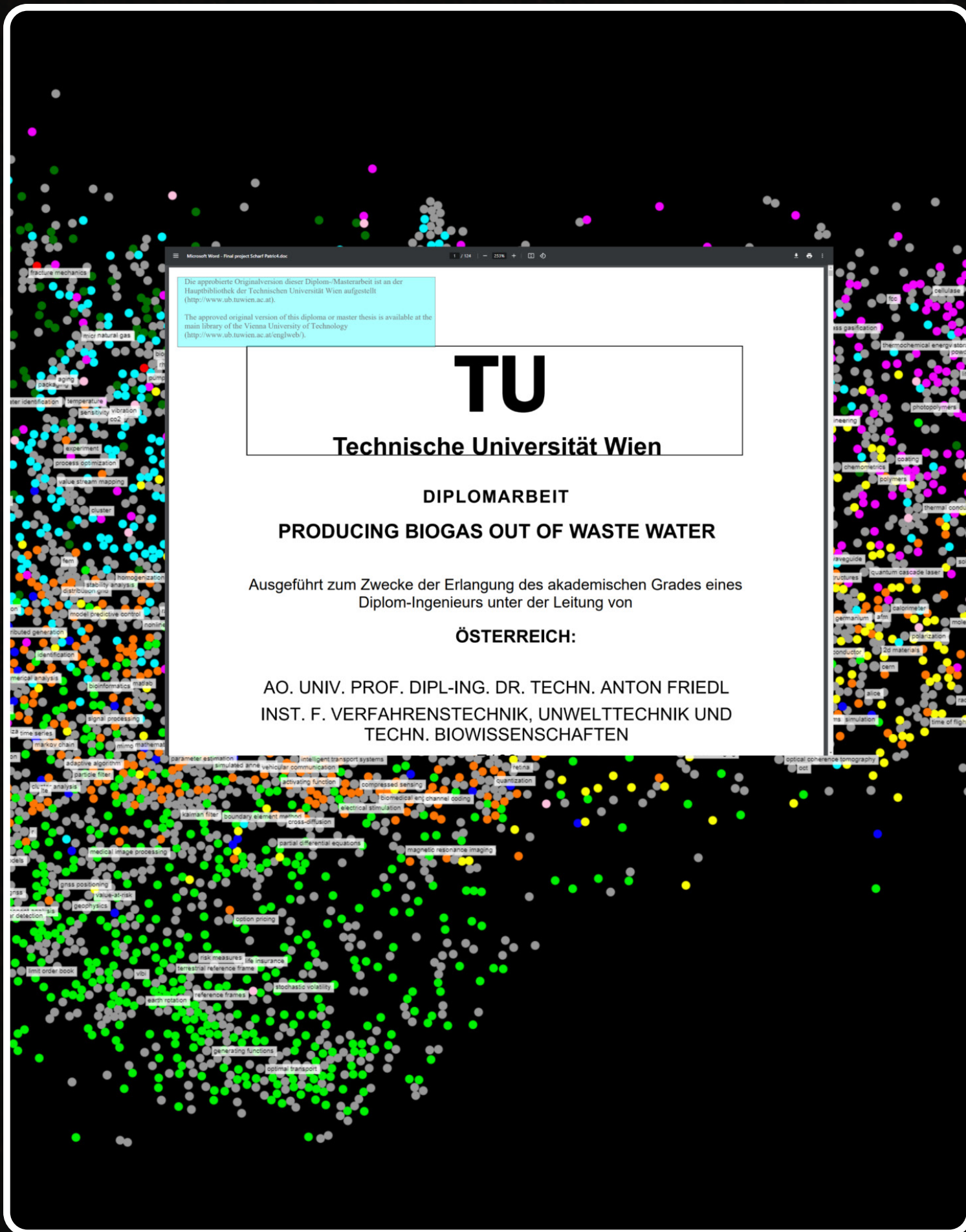


Bi-partite network of theses & keywords



VISUAL ENCODING STRATEGY

The node-edge nature of the data is directly translated into a visualization of the network's topology featuring 10,000 theses and an approximately equal amount of keywords. The layout has been created using the Force-Atlas-2 algorithm from Gephi. Keyword-nodes are uniformly shown in grey, while thesis-nodes are categorically colored by faculty. In order to prevent visual occlusion, labels are shown only for the most popular keywords. When a thesis-node is touched by the user, all its connected keyword labels are activated and a tooltip shows the title of the work. Double-touching a thesis-node enables users to read the thesis via a pop-up PDF-reader.



CONCLUSION

The two dashboard prototypes for the Davis visualization environment demonstrate the potential of creating compelling casual information visualizations for academic communities. Preliminary results from informal user tests suggest that the prototypes succeed in attracting users to playfully explore their personal academic interests and networks in a casual way. Future development will focus on broader usability and accessibility optimization, bug fixing and iterative functional expansion. These developments strive to position the Davis dashboards as exemplars of how interactive visualization can transform bibliographic datasets into meaningful, accessible, and inspiring experiences.

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